

H2O Online

Pune City Launch & Business Use Case

How to launch, manage, and scale a water delivery business in Pune, Maharashtra using the H2O Online platform — covering demand & supply management, stock & inventory, order workflows, admin operations, financial model, risks, and a 90-day rollout plan.

Covers:

- Phased Pune rollout (Kothrud → Baner → Wakad / Hinjewadi)
- Stock, demand & supply balancing workflows
- Supplier onboarding, rider fleet & store management
- 15 risks with mitigation controls
- Daily admin ops routine & KPI targets

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Workflow Diagrams

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Version: 1.0**City focus:** Pune, Maharashtra, India**Platform:** H2O Online Mobile + Admin Portal**Generated:** 29 June 2026

H2O Online — Business Use Case: Pune City Launch & Operations

Document type: Business Operations & Go-to-Market Guide

Version: 1.0

City focus: Pune, Maharashtra, India

Platform: H2O Online (Mobile App + Admin Portal + Backend API)

Production stack: AWS EC2 · MongoDB Atlas · Groq AI · Google Maps · Razorpay

1. Executive Summary

H2O Online is a **multi-sided water delivery platform** that connects households, societies, and businesses in Pune with verified water suppliers and delivery riders. The platform is not only a customer app — it is an **operations engine** for running a city-scale water delivery business: onboarding vendors, defining service zones by PIN code, tracking stock at store level, matching demand to supply, fulfilling orders end-to-end, and settling payments with commission-based economics.

Pune opportunity: Pune Metropolitan Region has 7M+ residents, rapid growth in IT corridors (Hinjewadi, Baner, Wakad), high apartment density, and recurring demand for **20L jar water**, **bulk cans**, and **tanker supply** in areas with irregular municipal supply. Existing delivery is fragmented (local vendors, phone orders, WhatsApp). H2O Online digitises discovery, ordering, tracking, subscriptions, and partner payouts in one system.

Business proposition for a Pune operator:

Stakeholder	Value
Platform owner / franchise	Commission on GMV, subscription retention, data on demand hotspots
Water supplier	New digital customers, order queue, rider fleet tools, wallet settlements
Delivery rider	Assigned jobs, GPS routes, 10% delivery share on order value
Customer	One app for instant + scheduled delivery, wallet, tracking, subscriptions
Admin team	Single dashboard for Pune operations — verify partners, zones, stock oversight, support

This document explains **how to launch and manage** H2O Online in Pune: workflows, stock/demand/supply management, admin roles, and risks.

2. Pune Market Context

2.1 Why Pune

Factor	Pune relevance
Population	~7M metro; large renter and young professional base
Housing	High-rise societies in Kothrud, Baner, Wakad, Hinjewadi, Magarpatta
Water need	Jar/can delivery standard; tanker demand in peripheral areas
Digital readiness	Strong smartphone penetration; UPI/wallet adoption
Competition	Local vendors dominate; no single city-wide digital leader

2.2 Target Customer Segments (Phase 1–3)

Segment	Pune examples	Platform feature
Households	Families in Kothrud, Aundh, Viman Nagar	Instant order, saved addresses, wallet
Societies / RWAs	Large complexes in Baner, Wakad	Society registration, bulk plans
Offices / startups	Hinjewadi IT parks, Kharadi	Scheduled orders, subscription billing
PG / co-living	Areas near colleges (VIT, MIT corridor)	Recurring subscription plans

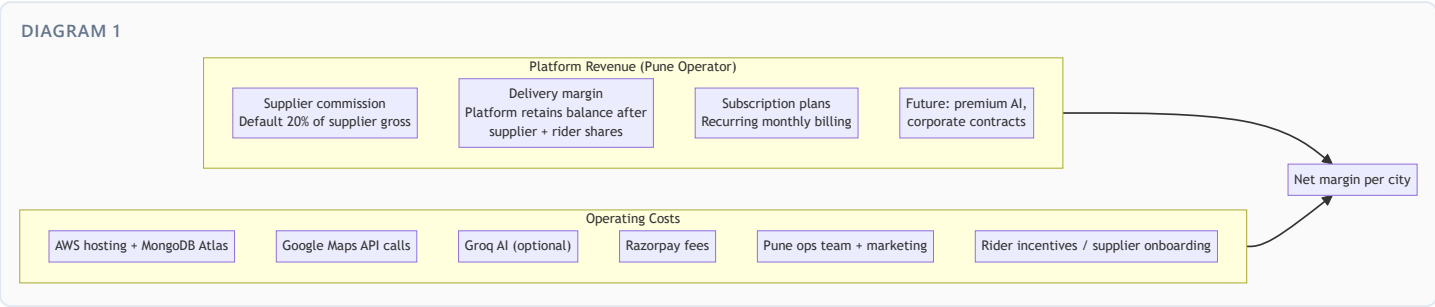
2.3 Suggested Pune Launch Zones (Phased)

Phase	Areas	Sample PIN codes	Rationale
Pilot (Month 1–2)	Kothrud, Erandwane, Karve Nagar	411038, 411004, 411052	Dense residential, manageable radius
Expand (Month 3–4)	Baner, Balewadi, Aundh	411045, 411007	High purchasing power
Scale (Month 5–6)	Wakad, Hinjewadi, Tathawade	411057, 411033	IT workforce, high order frequency

Phase	Areas	Sample PIN codes	Rationale
Later	Hadapsar, Magarpatta, Kharadi	411028, 411014	Larger geography, more riders needed

Admin configures these via **Serviceable Areas** (`/api/admin/serviceable-areas`) — PIN + radius (default 10 km) per supplier.

3. Business Model



3.1 Unit Economics (Per Order — Platform Defaults)

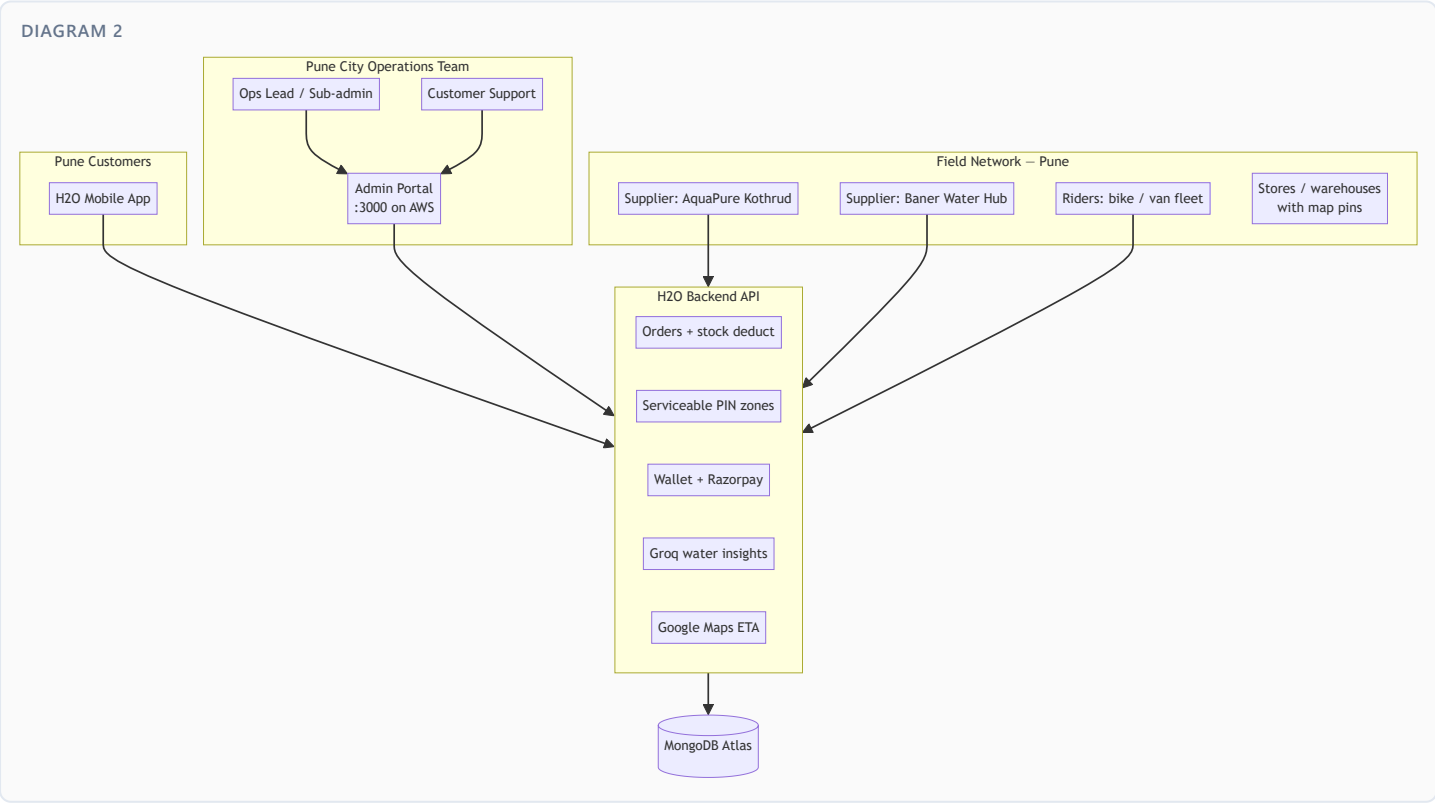
Party	Share	How platform calculates
Supplier	~80% of supplier item gross (configurable per supplier)	<code>commissionPercentage</code> on Supplier model; default 20% platform cut
Rider	10% of total order value (incl. tax)	<code>DELIVERY_SHARE_PERCENT = 10</code> in <code>orderPayout.js</code>
Platform	Remainder + commission	Credited on Razorpay/COD; debited from platform wallet on wallet-paid orders

Example: Customer in Kothrud orders 2 × 20L jars @ ₹80 each = ₹160 subtotal + tax → total ₹188.

- Rider share: ~₹19 (10%)
- Supplier payout: ~₹128 (80% of ₹160)
- Platform: commission ₹32 + any delivery margin after rider share

Settlements run automatically on **delivery confirmation** via `runPayoutOnDelivered()` .

4. Platform as Pune Operations Engine



What the admin portal manages for Pune:

Module	Business use in Pune
Suppliers	Onboard local Pune vendors; verify GST, bank, documents
Stores	Approve depot locations (Kothrud shop, Wakad warehouse)
Serviceable Areas	Map PIN codes to suppliers — control where orders are allowed
Products	City-wide catalogue oversight; ensure pricing consistency
Orders	Monitor live Pune order queue; intervene on failures
Delivery Partners	Verify riders; filter by vehicle (bike for jars, van for bulk)
Subscriptions & Plans	Family / office monthly plans for Pune societies
Wallet & Financials	Reconcile platform, supplier, rider balances
Tax Settings	Maharashtra GST configuration
Support	Handle Pune customer/supplier/rider tickets

5. Pune Launch Strategy

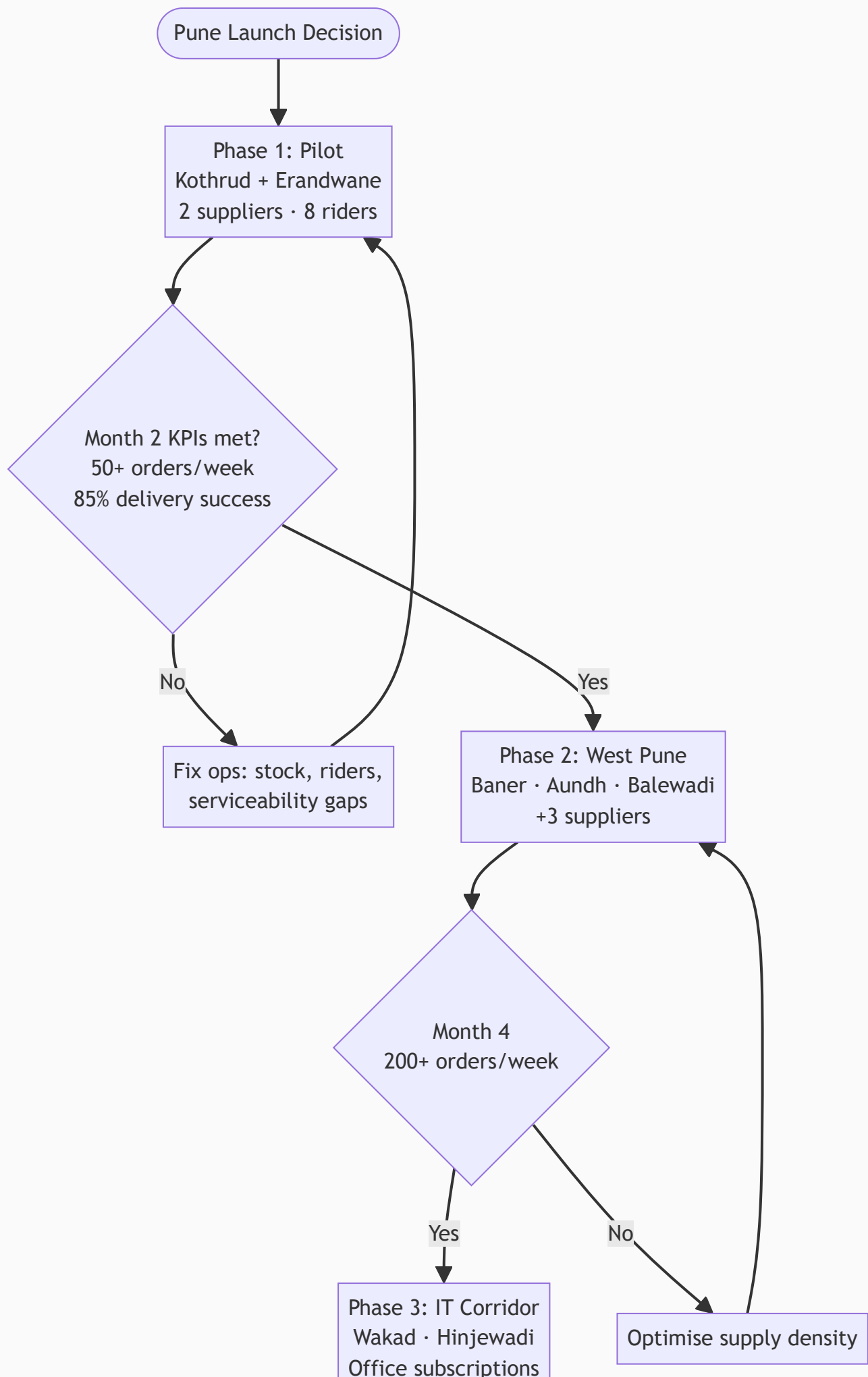
5.1 Pre-Launch Checklist (Weeks –4 to 0)

#	Task	Owner	Platform action
1	Sign 2–3 Pune water suppliers (pilot zone)	Business	Admin → Create/verify suppliers
2	Register supplier stores with GPS pins	Supplier + Admin	POST /api/stores → Admin approve
3	Define serviceable PINs for pilot	Ops	Admin → Serviceable Areas for 411038, 411004
4	Load products with stock quantities	Supplier	Products with stockQty , inStock
5	Recruit 5–10 riders (bike)	Ops	Register + admin verify delivery partners
6	Configure tax (GST) and Razorpay	Finance	Admin → Tax Settings; Razorpay live keys
7	Soft launch with 50 beta households	Marketing	Invite codes; monitor /api/admin/orders
8	Train supplier on accept/assign flow	Ops	Supplier app: incoming orders → assign rider

5.2 Phased Rollout Diagram

DIAGRAM 3





Phase 4: City-wide Pune
Hadapsar · Kharadi · Magarpatta

Managed city operation

5.3 Launch KPIs (Pune Pilot)

KPI	Target (Month 1)	Target (Month 6)	Where to measure
Active customers	100	2,000	Admin → Users
Weekly orders	50	800	Admin → Orders
Verified suppliers	2	12	Admin → Suppliers
Active riders	8	60	Admin → Delivery Partners
Order fulfilment rate	85%	95%	Delivered / total orders
Avg delivery time	< 45 min	< 35 min	Order tracking + Maps ETA
Subscription conversion	5%	20%	Admin → Subscriptions

6. Demand Management (Pune)

Demand = customer orders + subscriptions + seasonal spikes (summer, festivals, office reopenings).

DIAGRAM 4

Demand Sources – Pune

Instant orders
App → Checkout

Scheduled orders
orderType: scheduled

Subscriptions
Monthly plans + bills

Society bulk
Corporate channel



Demand Signals (Admin monitors)

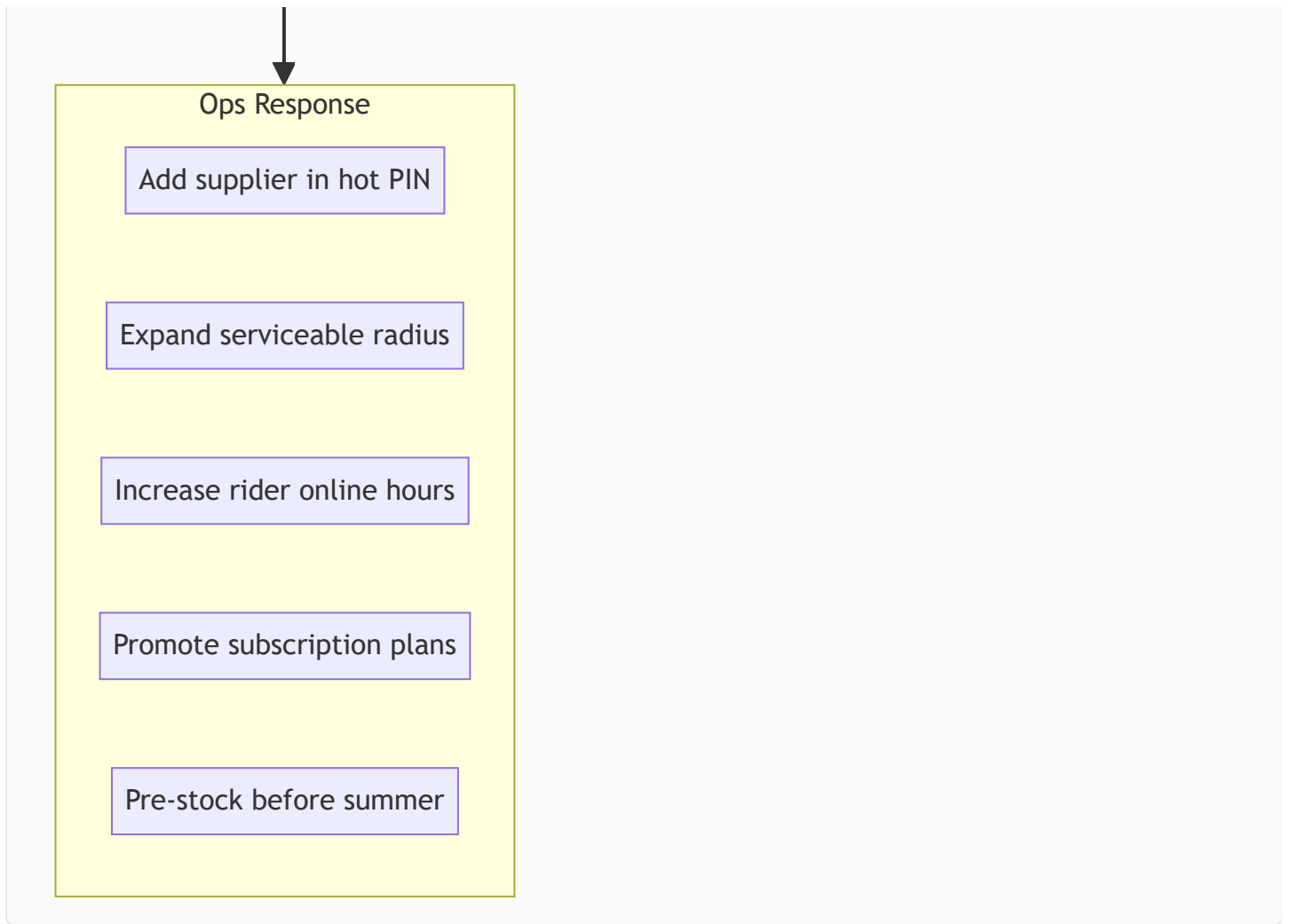
Orders by PIN / area

Peak hours 7-9 AM, 6-8 PM

Product capacityL mix
20L jar vs bulk

Rejected orders
supplier capacity

Unserviceable checkout blocks



6.1 Managing Demand by Product Type

Product type	Pune use case	Demand pattern	Platform handling
20L jar	Household daily	High morning/evening	Instant order; stock per store
5L / 10L bottles	Small families	Moderate	Product catalog filters
Bulk / tanker	Societies, events	Sporadic high volume	vehicleType: tanker riders
Subscription plan	RWAs, offices	Predictable monthly	Plans → Subscriptions → Bills

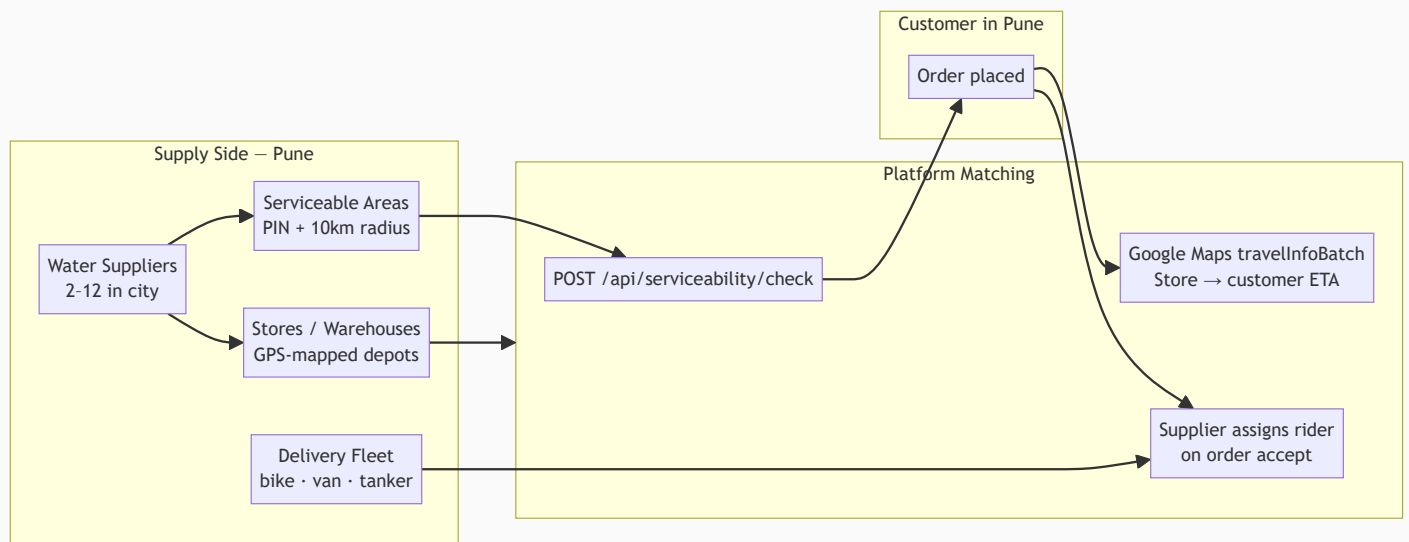
6.2 Seasonal Demand (Pune)

Season	Demand impact	Ops playbook
Summer (Mar-Jun)	+40–60% jar orders	Pre-load stockQty ; add riders; extend service hours
Monsoon	Delivery delays	Increase ETA buffer; rider safety SOP
Festivals (Ganesh, Diwali)	Spike + bulk	Temporary serviceable areas; society campaigns
IT campus cycles	Hinjewadi weekday peaks	Schedule rider shifts 11 AM–2 PM, 6–9 PM

7. Supply Management (Pune)

Supply = verified suppliers + approved stores + online riders + serviceable geography.

DIAGRAM 5



7.1 Supplier Onboarding (Pune Local Vendors)

Step	Business action	Platform workflow
1	Identify Pune jar/tanker vendor	Field sales visit
2	Vendor registers on app	POST /api/auth/register-supplier
3	Submit GST, bank, license docs	Supplier onboarding screen
4	Admin verifies	Admin → Suppliers → Verify
5	Vendor adds store with map pin	POST /api/stores → Admin approve
6	Admin maps serviceable PINs	Serviceable Areas for vendor
7	Vendor lists products + stock	Products with stockQty
8	Go live in zone	Orders appear in supplier incoming queue

Target supplier density: 1 supplier per 3–5 km² in pilot; avoid overlap conflicts unless intentional for competition.

7.2 Rider Fleet Management

Rider type	Pune deployment	Vehicle in platform
Jar delivery	Kothrud, Baner lanes	bike, bicycle
Multi-jar / office	Aundh, Viman Nagar	van, miniTruck
Society tanker	Peripheral Pune	tanker

Three rider onboarding paths:

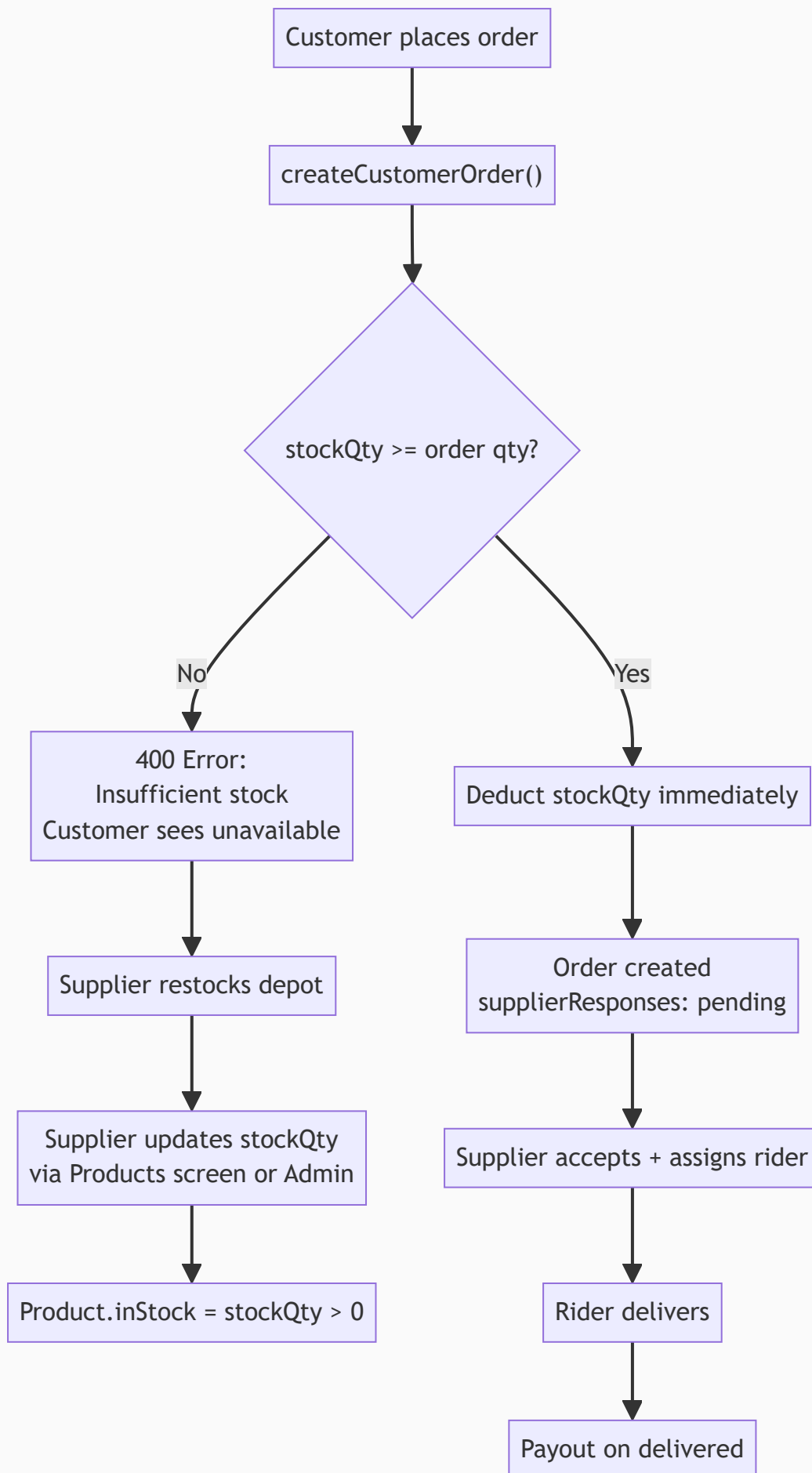
1. **Self-register** → Admin verify (city-wide gig riders)
2. **Supplier fleet** → Supplier adds via POST /api/supplier/delivery-partners (instant, managed)
3. **Supplier-as-rider** → Single-owner vendors who deliver themselves

Availability rules: Rider must be approved, online, and not on another active delivery (partnerAvailability.js).

8. Stock & Inventory Management

The platform tracks stock at **product level per supplier/store**. Stock is **deducted at order creation** (not at delivery) — critical for Pune ops planning.

DIAGRAM 6



8.1 Stock Operations Playbook (Pune)

Activity	Frequency	Who	How
Set opening stock	Daily 6 AM	Supplier	Update stockQty per product per store
Monitor low stock	Continuous	Admin / Supplier	Admin → Products; filter low stock
Reorder from plant	When stock < 2 days cover	Supplier ops	Off-platform; reflect in app
Multi-store allocation	Per area	Supplier	Link products to storeId (Kothrud store vs Wakad warehouse)
Prevent overselling	Automatic	Platform	Server rejects order if insufficient stock
Cancelled order stock	On cancel	*Gap today*	Manual restock adjustment by supplier

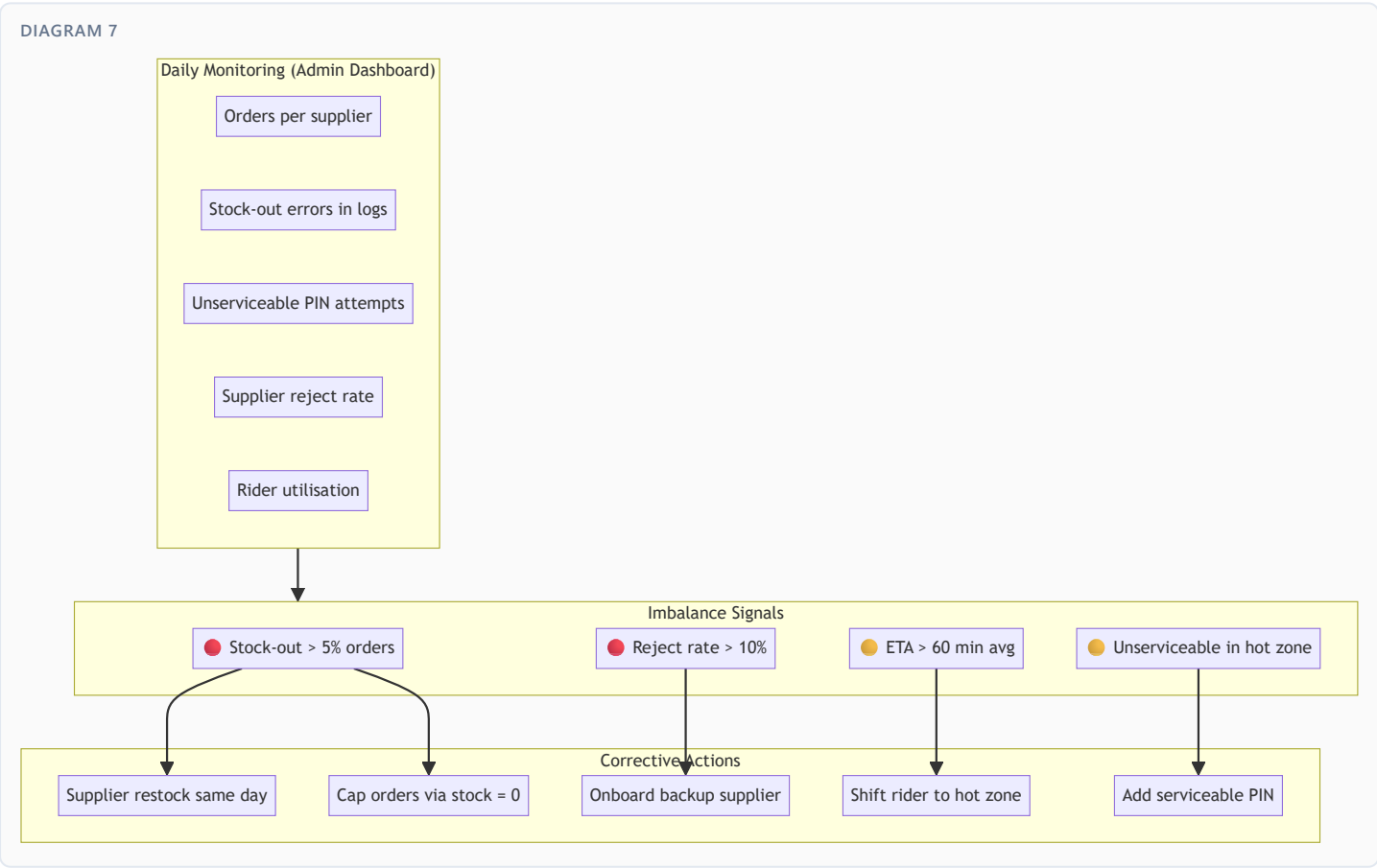
8.2 Stock Planning Formula (Pune Jar Business)

Daily stock needed (jars) =
(Avg daily orders × avg jars per order × 1.15 safety factor)
+ subscription scheduled deliveries

Example — Kothrud depot:

Input	Value
Avg daily orders	40
Avg jars per order	2
Safety factor	1.15
Daily stock	40 × 2 × 1.15 = 92 jars
Opening stock target	100 jars by 6 AM

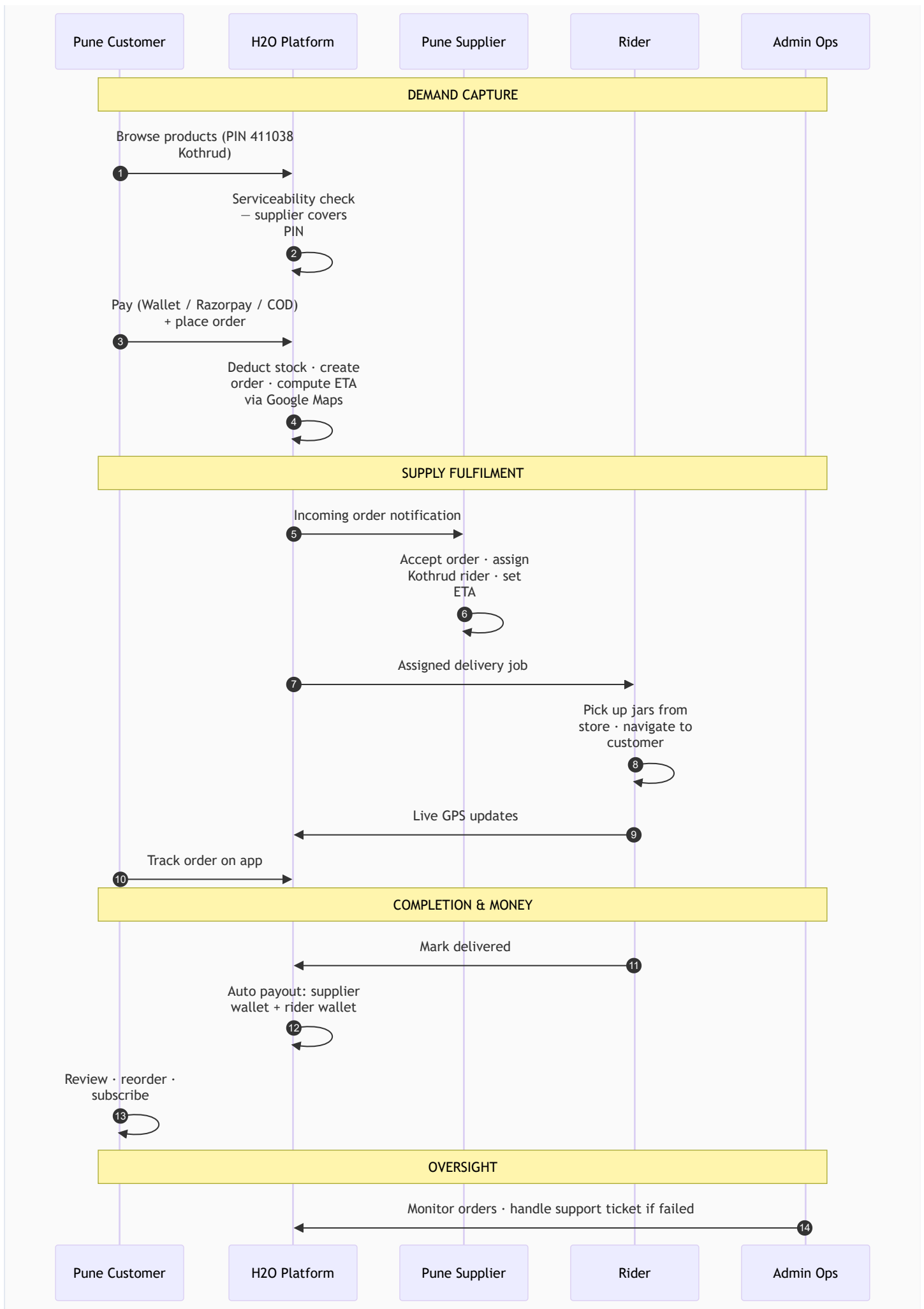
8.3 Demand–Supply Balance Workflow



9. Order Fulfilment — Business Workflow

End-to-end business workflow connecting Customer → Platform → Supplier → Rider → Settlement.

DIAGRAM 8



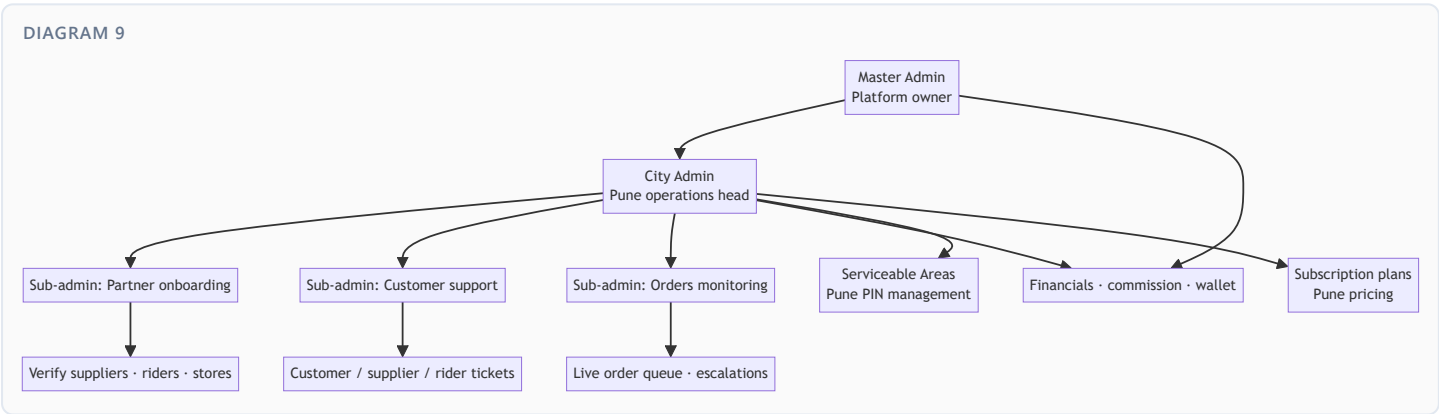
9.1 Order States — Business Meaning

Platform state	Business meaning for Pune ops
in_progress + supplier pending	Order waiting — supplier must accept within SLA (target: 5 min)
accepted + rider assigned	Fulfillment in progress — track ETA
picked_up	Rider has stock — customer expects delivery soon
delivered	Complete — payout triggered
cancelled	Lost revenue — analyse reason (stock, rider, customer)
rejected by supplier	Supply failure — reassign zone or supplier

9.2 SLA Targets (Pune Operations)

Stage	Target SLA	Escalation
Supplier accept	5 minutes	Admin calls supplier
Rider assignment	2 minutes after accept	Auto-suggest fleet rider
Pickup	15 minutes after assign	Rider support
Delivery (instant)	45 minutes total	Customer support ticket
Delivery (scheduled)	±15 min of slot	SMS/app notification

10. Admin Roles — Pune City Management



10.1 Daily Admin Routine (Pune Ops Center)

Time	Activity	Admin module
7:00 AM	Check overnight orders; verify rider online count	Dashboard, Orders
8:00 AM	Review supplier stock levels	Products
9:00 AM–6:00 PM	Monitor live orders; resolve support tickets	Orders, Support
12:00 PM	Peak lunch check — rider availability	Delivery Partners
6:00 PM	Evening peak prep	Orders, Serviceable Areas
9:00 PM	Day close — financials snapshot	Financials
Weekly	Verify new supplier/rider applications	Suppliers, Delivery Partners
Monthly	Subscription bill review; commission adjustment	Subscriptions, Financials

10.2 Sub-admin vs Admin (Pune team structure)

Role	Pune team example	Permissions
Master	Founder / platform owner	Everything + master credentials
Admin	Pune city manager	Full ops including financials
Sub-admin	Support agent, onboarding exec	No financials, no delete user, no remove supplier

11. Subscriptions & Recurring Revenue (Pune)

Subscriptions stabilise demand — critical for **society RWAs** and **office parks** in Baner/Hinjewadi.

Step	Business action	Platform
1	Design Pune plans (Basic Family, Office 50L/day)	Admin → Plans
2	Customer selects plan on app	GET /api/plans
3	Customer subscribes	POST /api/subscriptions
4	Monthly bill generated	SubscriptionBill
5	Customer pays bill	POST /api/bills/:id/pay
6	Supplier/rider fulfil on schedule	Delivery partner subscription routes

Business benefit: Predictable demand → suppliers pre-load stock → fewer stock-outs → higher retention.

12. Financial Management (Pune Operation)

Flow	Description
Customer pays	Razorpay (online), Wallet (prepaid), COD (cash on delivery)
Platform holds	Razorpay settlement to platform bank account
On delivery	Auto-split to supplier wallet + rider wallet
Supplier withdraws	Off-platform bank transfer (manual today)
Platform revenue	Commission % + delivery margin; view in Admin → Financials
GST	Configured in Admin → Tax Settings (Maharashtra rates)

Wallet note: For wallet-paid orders, platform wallet is debited for payouts — ops must maintain sufficient platform wallet float.

13. Risks & Challenges (Pune Specific)

13.1 Risk Register

ID	Risk	Likelihood	Impact	Category
R01	Stock-out during peak (summer, morning rush)	High	High	Supply
R02	Insufficient riders in new zone	High	High	Supply
R03	Supplier rejects orders (capacity, offline)	Medium	High	Supply
R04	Wrong serviceability config — customer can't order	Medium	Medium	Ops
R05	Delivery delays — Pune traffic (Hinjewadi, JM Road)	High	Medium	Logistics
R06	Monsoon — waterlogging, rider safety	High	Medium	Seasonal
R07	Local vendor resistance — won't join platform	Medium	Medium	Business
R08	Price undercutting by offline vendors	High	Medium	Competition
R09	Quality complaints — water purity disputes	Medium	High	Trust
R10	Payment failures — Razorpay downtime	Low	Medium	Tech
R11	Google Maps API cost at scale	Medium	Low	Tech
R12	Regulatory — FSSAI, local trade license	Medium	High	Compliance
R13	Multi-supplier order complexity	Low	Medium	Platform
R14	Cancelled order stock not auto-restored	Medium	Low	Platform gap
R15	Data/connectivity — rider GPS gaps	Medium	Medium	Tech

13.2 Challenge Deep-Dives

Challenge 1 — Morning demand spike (7–9 AM)

Pune households order jars before work. If stock is updated late, first 50 customers may fail checkout.

Mitigation: Mandatory 6 AM stock update SOP; admin dashboard alert if any product `stockQty < 20`.

Challenge 2 — Geographic expansion faster than supply

Launching Wakad before having a Wakad depot causes unserviceable errors and bad reviews.

Mitigation: Never enable PIN in Serviceable Areas until store is approved within radius.

Challenge 3 — Supplier commission negotiation

Pune vendors may resist 20% commission vs WhatsApp orders at 0%.

Mitigation: Phase 1 lower commission (10–15%); volume guarantee; subscription lead-sharing.

Challenge 4 — Rider attrition

Gig riders switch to food delivery apps.

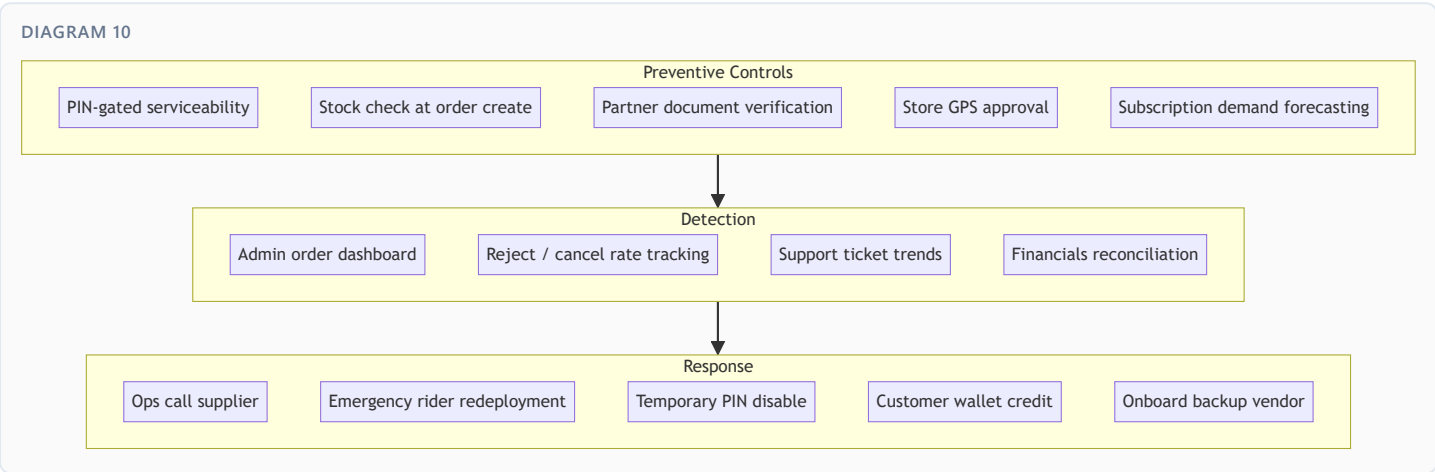
Mitigation: Supplier-managed fleet in core zones; delivery share visibility in app; peak incentives.

Challenge 5 — Trust in new digital brand

Customers prefer existing "Bhaiya" vendor.

Mitigation: Verified badges; society RWA partnerships; first-order wallet credit.

14. Mitigation & Controls



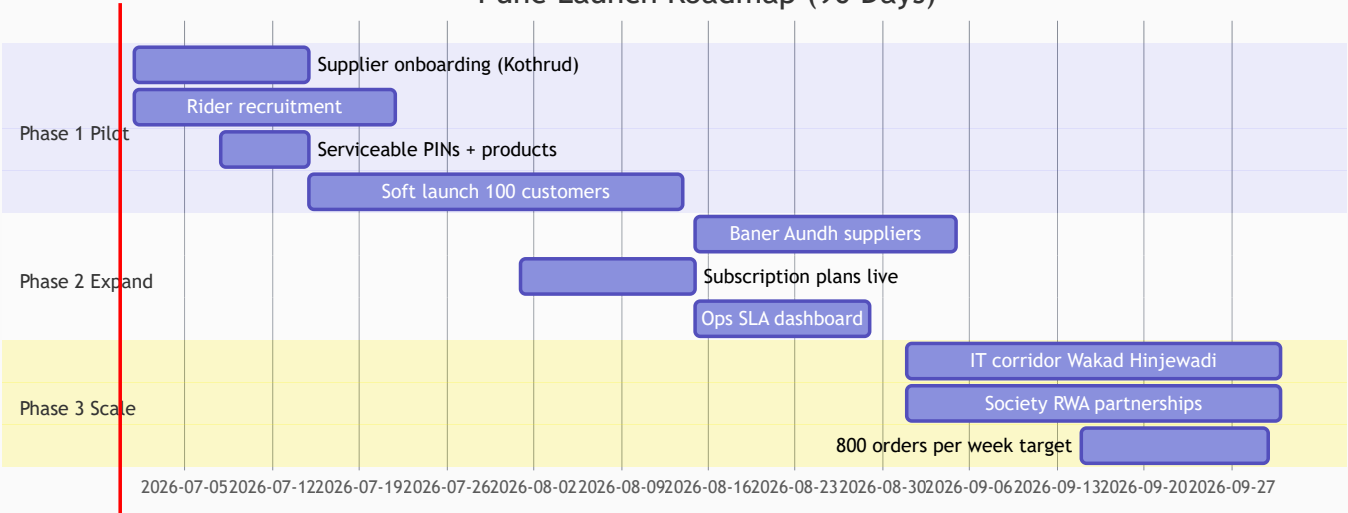
Control	Platform feature	Pune ops action
Prevent overselling	stockQty deduct on order	Daily stock SOP
Zone control	Serviceable Areas	Don't expand PIN without supply
Partner quality	Admin verify suppliers/riders	Field audit before approve
Payment security	Razorpay HMAC verify	Use live keys in production
Customer recovery	Support tickets + wallet credit	Same-day resolution SLA

15. 90-Day Pune Launch Roadmap

Period	Focus	Key deliverables
Days 1–30	Pilot setup	2 suppliers, 8 riders, Kothrud PINs, 100 customers, 50 orders/week
Days 31–60	Optimise ops	Stock SOP, SLA monitoring, subscription plan launch, Baner expansion
Days 61–90	Scale	Wakad/Hinjewadi, 12 suppliers, 60 riders, 800 orders/week, society partnerships

DIAGRAM 11

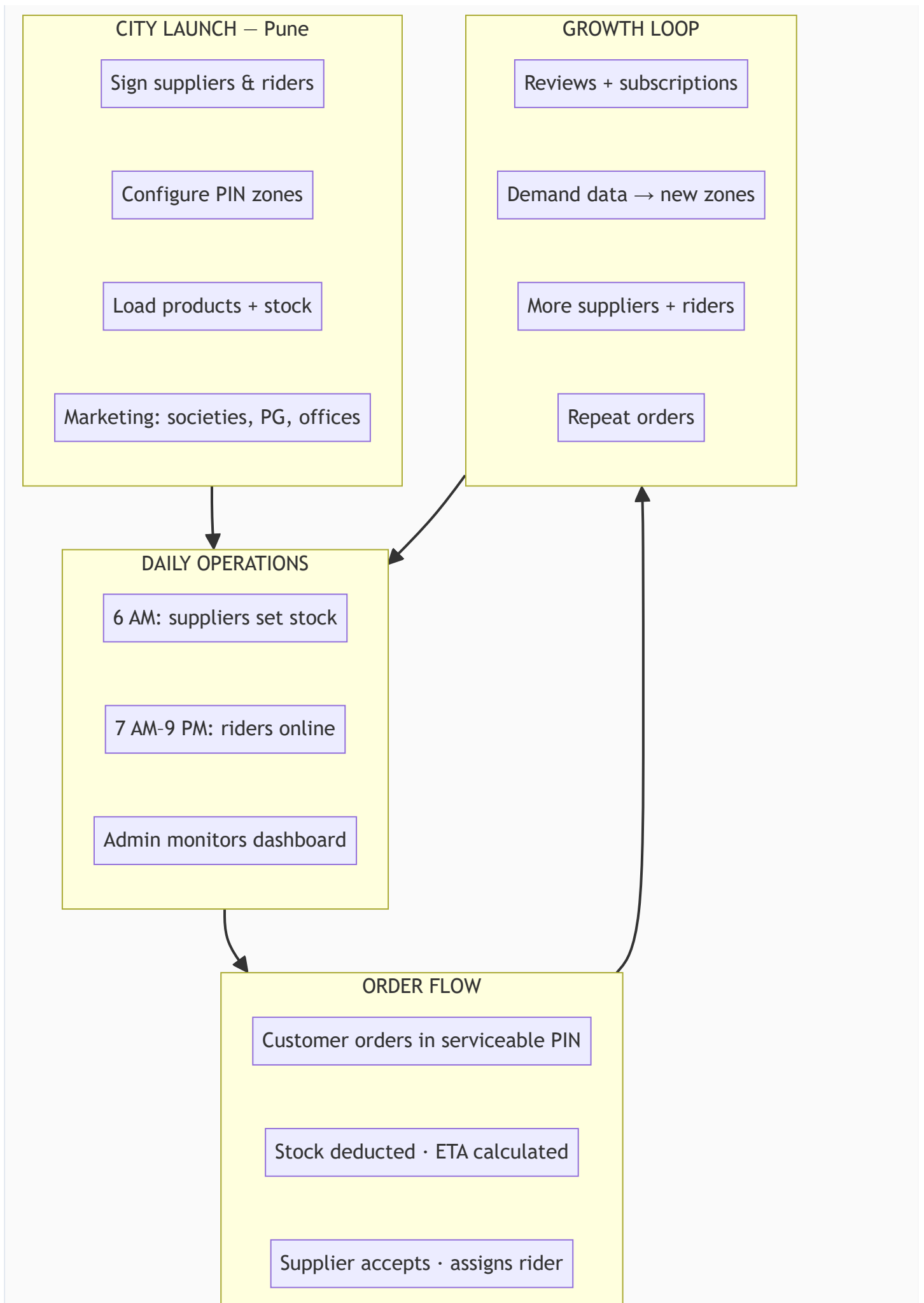
Pune Launch Roadmap (90 Days)



16. Complete Business Use Case Workflow

Single diagram linking **launch** → **daily ops** → **order** → **money** → **growth**.

DIAGRAM 12



Delivered · payout split

17. Success Criteria — Pune City Operation

Dimension	6-month success definition
Market	Top-3 digital water app in pilot zones (Kothrud, Baner, Wakad)
Volume	800+ orders/week; 200+ active subscriptions
Supply	12 verified suppliers; 95% fulfilment rate
Unit economics	Positive contribution margin per order after variable costs
Customer	4.2+ avg rating; < 5% support escalation rate
Platform	99% API uptime; stock-out rate < 3%

18. Related Platform Documentation

Document	Purpose
docs/SYSTEM-ARCHITECTURE.md	Technical architecture & data flows
docs/H2O-System-Architecture.pdf	Architecture PDF with diagrams
docs/DEPLOYMENT-GUIDE.md	AWS production deployment
docs/requirement-capture.md	Full functional requirements
config/aws-production.json	Production API URLs

This business use case is aligned with H2O Online platform capabilities as implemented in the codebase: stock deduction on order, serviceable PIN areas, supplier/rider verification, commission-based payouts, subscriptions, and admin portal operations.